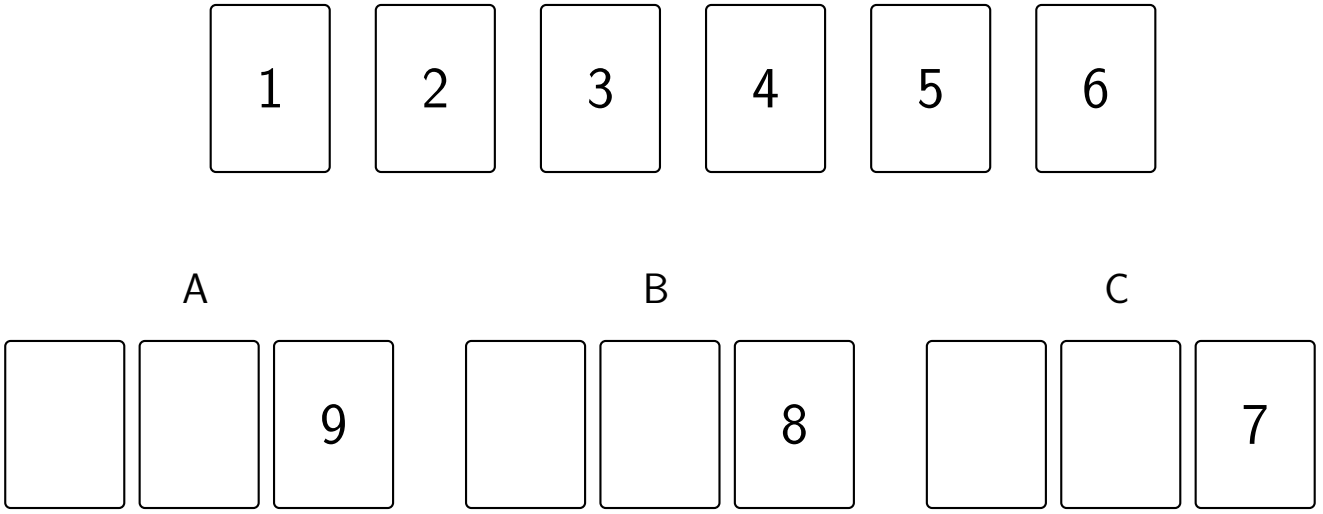
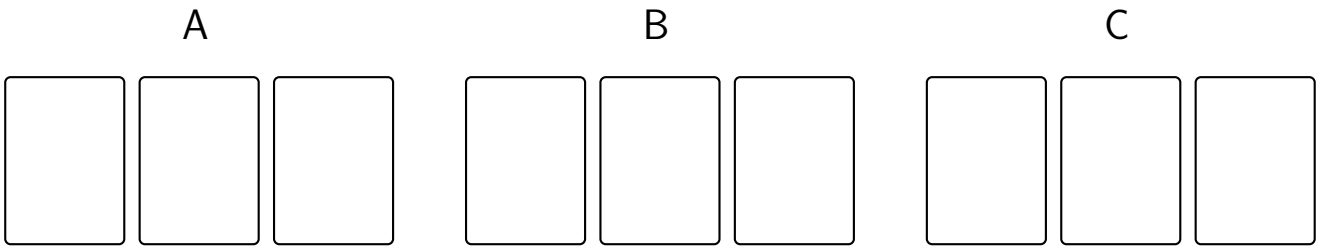
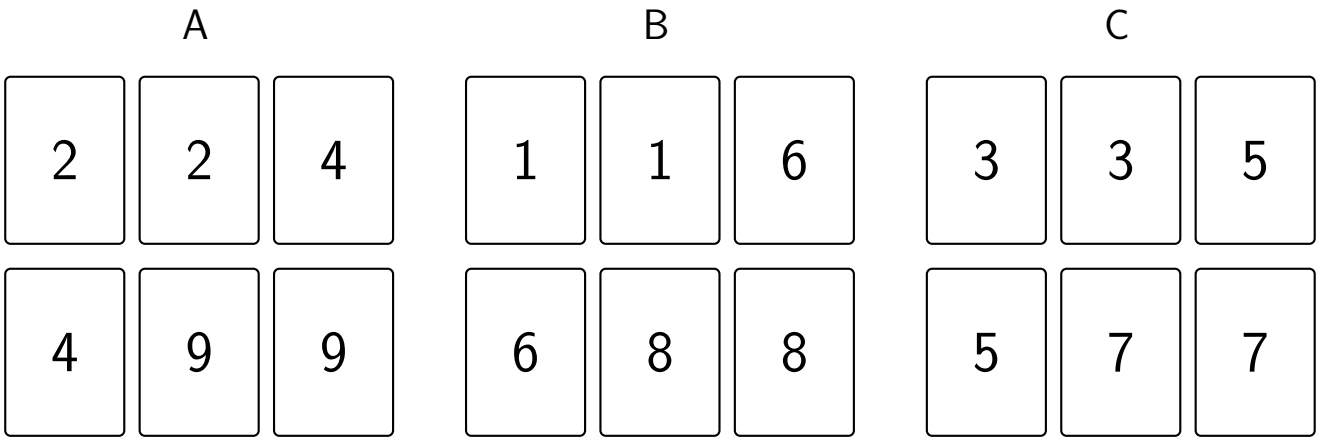
**Problem 2:** Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

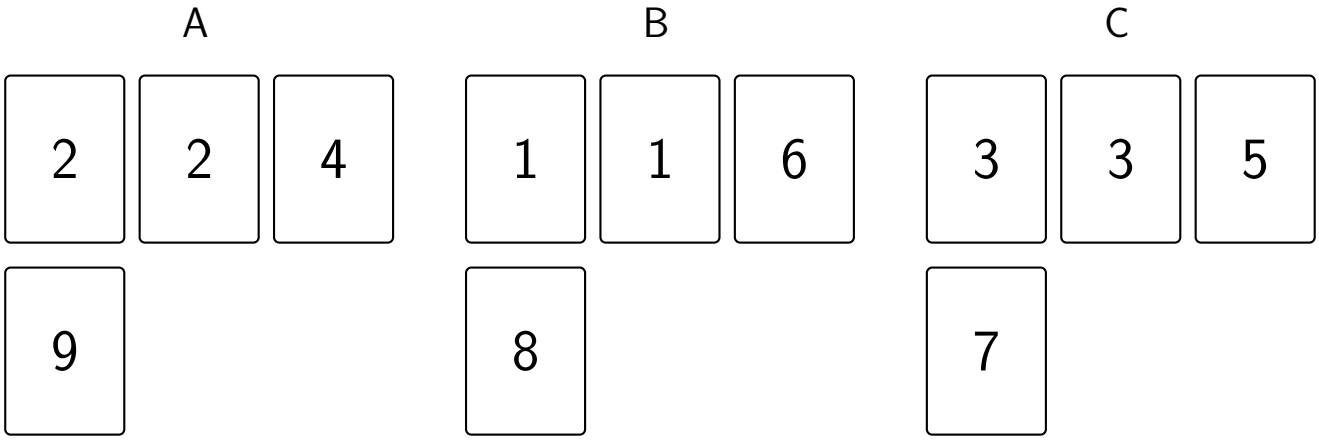


**Problem 3:** Write numbers from 1 to 6 on these nine cards. A number may repeat within one deck, but it cannot appear in two different decks. Make A win more than half its pairs against B, B against C, and C against A.

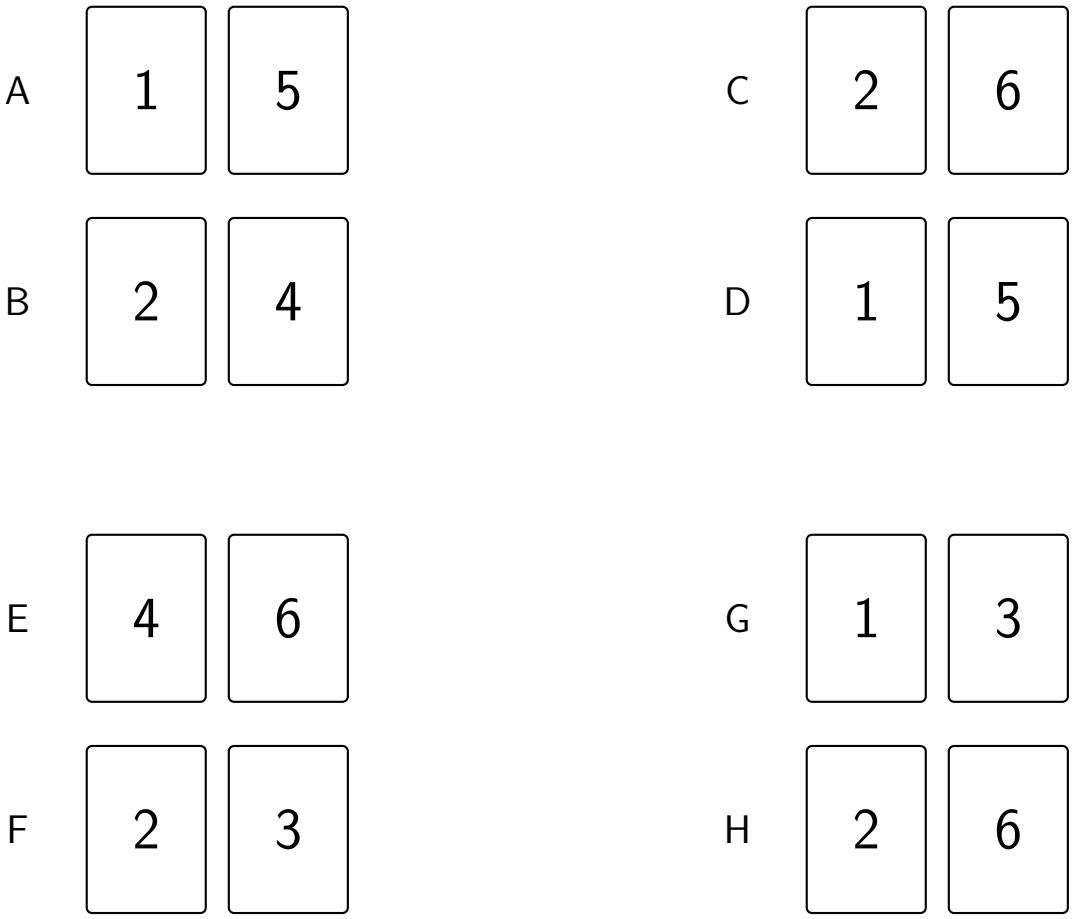


**Problem 4:** Find each deck's chance of winning against each other deck, first with the six-card decks and then with the four-card decks. Which chances changed from Problem 1?

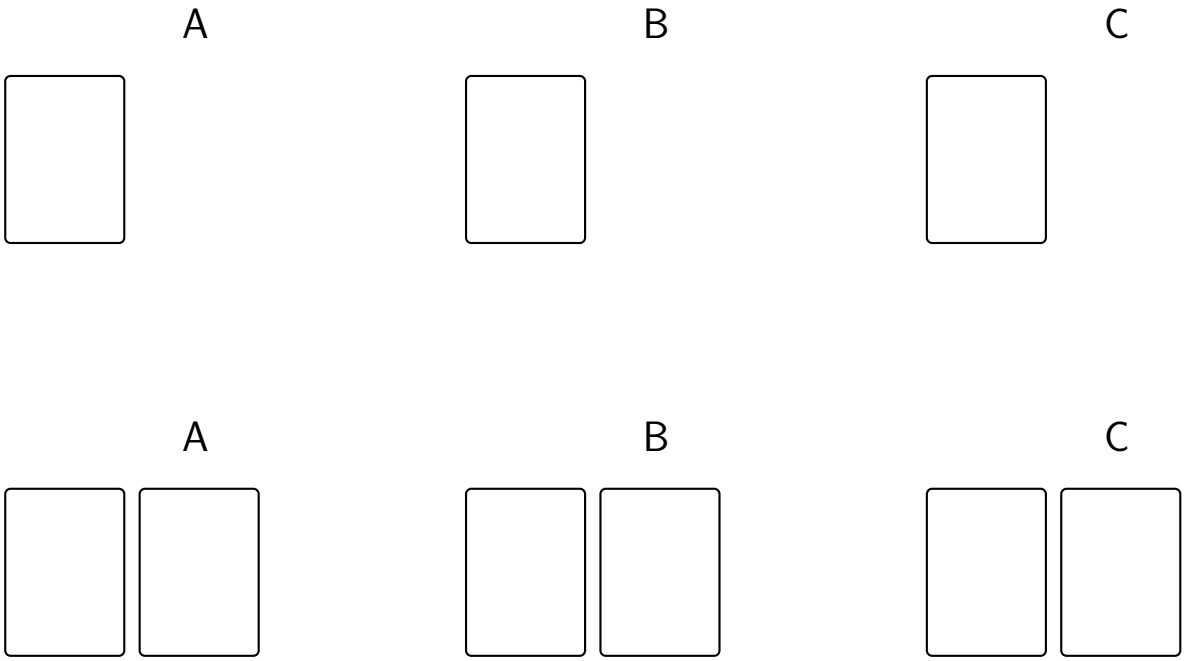


**Problem 5:** Decide which deck, if either, wins more than half the pairs in each comparison. Find a rule that works for every pair of two-card decks with no number shared between decks, and explain why it works.



**Problem 6:** Can three one-card decks make A win more than half its pairs against B, B against C, and C against A? Can three two-card decks do it? Explain for any choice of numbers. What is the fewest cards per deck that allows this, with all decks the same size?



**Problem 7:** Use cards 1 to 9 once each in three decks of three. Can A win at least six of its nine pairs against B, B at least six against C, and C at least six against A? Find an arrangement, or explain why it is impossible.

